
Integrable Systems — Exercise Sheet 7

When integrability fails

Variant: Questions only

Conventions, as in Lecture 7: in action–angle variables the angles φ are the positions and the actions I the momenta, and $\{f, g\} = \sum_i (\partial_{\varphi_i} f \partial_{I_i} g - \partial_{I_i} f \partial_{\varphi_i} g)$. For $H = H_0(I) + \varepsilon H_1(\varphi, I)$ with $H_1 = \sum_k h_k(I) e^{ik \cdot \varphi}$ and $\nu = \partial H_0 / \partial I$, the homological equation is $\nu \cdot \partial_{\varphi} S_1 = h_0 - H_1$, and Lecture 7's proposition on first-order perturbation theory solves it by $S_1 = \sum_{k \neq 0} i h_k e^{ik \cdot \varphi} / (k \cdot \nu)$, under its hypotheses (no $k \cdot \nu$ with $h_k \neq 0$ vanishes, and the series converges); the new Hamiltonian is then $H_0 + \varepsilon h_0 + O(\varepsilon^2)$. An oscillator $\frac{1}{2}(p^2 + \nu^2 q^2)$, $\nu > 0$, has $q = \sqrt{2I/\nu} \sin \varphi$, $p = \sqrt{2I\nu} \cos \varphi$ and $H_0 = \nu I$ (Lecture 6; Sheet 3, Exercise 4).

Exercise 1 (first order, with one degree of freedom). Let $H = \frac{1}{2}(p^2 + \nu^2 q^2) + \varepsilon q^4$ with $\nu > 0$. (i) Show that $H = \nu I + \varepsilon H_1$ with $H_1 = \frac{I^2}{\nu^2} (\frac{3}{2} - 2 \cos 2\varphi + \frac{1}{2} \cos 4\varphi)$, and find h_0 and the other h_k . (ii) Solve the homological equation. Which denominators $k\nu$ appear, and can any of them vanish? (iii) Deduce that, to first order in ε , the frequency is $\nu + 3\varepsilon I/\nu^2$. (iv) Take $\nu = 1$, $\varepsilon = 0.01$ and the energy $E = \frac{1}{2}$. Show that the first-order frequency at energy E is $1 + 3\varepsilon E + O(\varepsilon^2)$, and compare it with $2\pi/T$, the period $T = \oint dq/p$ computed numerically.

Exercise 2 (two oscillators and a resonant denominator). Let $H = \frac{1}{2}(p_1^2 + \nu_1^2 q_1^2) + \frac{1}{2}(p_2^2 + \nu_2^2 q_2^2) + \varepsilon q_1^2 q_2$ with $\nu_1, \nu_2 > 0$, and action–angle variables (φ_i, I_i) for each oscillator, $I_1, I_2 > 0$. (i) Show that $H_0 = \nu_1 I_1 + \nu_2 I_2$ and

$$H_1 = A \left(\frac{1}{2} \sin \varphi_2 - \frac{1}{4} \sin(2\varphi_1 + \varphi_2) + \frac{1}{4} \sin(2\varphi_1 - \varphi_2) \right), \quad A = \frac{2I_1}{\nu_1} \sqrt{\frac{2I_2}{\nu_2}}.$$

Which Fourier modes k occur, and what is h_0 ? (ii) Solve the homological equation. Which denominator can vanish, and for which frequencies? For those frequencies, which tori are resonant? Is H_0 non-degenerate? (iii) At $\nu_1 = \nu$, $\nu_2 = 2\nu$ and $\varepsilon \neq 0$, keep only H_0 and the resonant term: $H_{\text{res}} = \nu(I_1 + 2I_2) + \frac{1}{4}\varepsilon A \sin(2\varphi_1 - \varphi_2)$. Show that $I_1 + 2I_2$ is a first integral of H_{res} but I_1 is not.

Exercise 3 (the golden ratio is Diophantine). Let $g = (\sqrt{5} - 1)/2$, and recall from Lecture 7 that $\nu \in \mathbb{R}^2$ is (γ, τ) -Diophantine if $|k \cdot \nu| \geq \gamma/|k|^\tau$ for every $k \in \mathbb{Z}^2 \setminus \{0\}$, where $|k| = |k_1| + |k_2|$. (i) Show that g and $g' = -(\sqrt{5} + 1)/2$ are the roots of $x^2 + x - 1$, and that for integers p, q , $(qg - p)(qg' - p) = p^2 + pq - q^2$. (ii) Deduce that $|qg - p| \geq 1/((1 + \sqrt{5})q)$ for all integers p and $q \geq 1$. (iii) Deduce that $\nu = (1, g)$ is $(\gamma, 1)$ -Diophantine with $\gamma = 1/(1 + \sqrt{5})$. Why is no ν with a rational ratio ν_2/ν_1 Diophantine, for any $\gamma > 0$? (iv) (Computational.) Compute $m(N) = \min_{0 < |k| \leq N} |k| |k \cdot \nu|$ for $\nu = (1, g)$ and $\nu = (1, \pi)$, at $N = 10, 100$ and 1000 . What do the numbers say?

Exercise 4 (twist maps and the standard map). (i) For the rotors $H_0 = \frac{1}{2}(I_1^2 + I_2^2)$ of Lecture 7, on the level $H_0 = E$ with the section $\varphi_2 = 0$ and $I_2 > 0$, find the rotation number $\rho(I_1)$ of the unperturbed section map, and show that it is a twist map. Show that for the isotropic oscillator, $H_0 = I_1 + I_2$, which Hénon–Heiles perturbs, the unperturbed section map is the identity. (ii) Show that the standard map $p' = p + K \sin x$, $x' = x + p'$ preserves area. (iii) For $K > 0$, find its fixed points on the cylinder (x modulo 2π) and classify them. With M the

Jacobian, $\det M = 1$, a fixed point is *hyperbolic* if $|\operatorname{tr} M| > 2$ (real eigenvalues $\mu, 1/\mu$ with $|\mu| > 1$), *elliptic* if $|\operatorname{tr} M| < 2$ (eigenvalues $e^{\pm i\alpha}$), and *parabolic* if $|\operatorname{tr} M| = 2$. (iv) For a map, the Lyapunov exponent is $\lambda = \lim_{m \rightarrow \infty} \frac{1}{m} \ln (|DP^m u|/|u|)$. At $K = 1$, find λ at the hyperbolic fixed point, for u not along the contracting eigenvector. Is this positive exponent a sign of chaos?

Exercise 5 (Hénon–Heiles and the Toda ring). Let $V = \frac{1}{2}(x^2 + y^2) + x^2y - \frac{1}{3}y^3$, the Hénon–Heiles potential. (i) Show that in polar coordinates $V = \frac{1}{2}r^2 + \frac{1}{3}r^3 \sin 3\theta$. Which symmetries does this show? (ii) Find all critical points of V , their type and their heights. (iii) Show that the Toda ring’s potential of Lecture 7, $\frac{1}{24}(e^{2y+2\sqrt{3}x} + e^{2y-2\sqrt{3}x} + e^{-4y}) - \frac{1}{8}$, is $V + \frac{1}{2}(x^2 + y^2)^2 + O(r^5)$. (iv) In the family $V_\lambda = \frac{1}{2}(x^2 + y^2) + x^2y + \frac{\lambda}{3}y^3$ (Hénon–Heiles is $\lambda = -1$), show that $\lambda = 1$ separates in $u = (x + y)/\sqrt{2}$, $v = (y - x)/\sqrt{2}$, and write down a second integral.

Exercise 6 (sections and a Lyapunov exponent, computed). (i) Write the section code of Lecture 7’s tutorial and reproduce the Hénon–Heiles sections $x = 0$, $p_x > 0$ at $E = \frac{1}{12}, \frac{1}{8}$ and $\frac{1}{6}$, checking that H drifts by less than 10^{-8} along each orbit. (ii) Do the same for the Toda ring of Exercise 5 at $E = \frac{1}{6}$. What do you see? (iii) Estimate the largest Lyapunov exponent of the Hénon–Heiles orbits launched on the section at $(y, p_y) = (0, 0)$, $(-0.1, 0)$ and $(0.3, 0)$ at $E = \frac{1}{6}$, and at $(0, 0)$ at $E = \frac{1}{12}$: integrate the linearised equations along the orbit from a generic unit tangent vector (try two), renormalise it every 10 time units, and average the logarithms of the growth factors, up to $t = 5 \times 10^4$. How does the estimate depend on the launch point, the starting vector and the length t of the run? What goes wrong if the starting vector is $(1, 0, 0, 0)$?

Exercise 7 ★. Poincaré’s mechanism, tested. Take Lecture 7’s rotors with $V = \cos q_1 \cos q_2$: $H = \frac{1}{2}|p|^2 + \varepsilon \cos q_1 \cos q_2$ on $T^2 \times \mathbb{R}^2$, with q the angles and $p = I$, so that $H_0 = \frac{1}{2}|I|^2$ and $\nu = I$. (i) Find the Fourier coefficients h_k of V and the Poincaré set $\mathfrak{P}$. Is it dense? (ii) Show that $F = \frac{1}{4}(p_1 + p_2)^2 + \frac{1}{2}\varepsilon \cos(q_1 + q_2)$ and $G = \frac{1}{4}(p_1 - p_2)^2 + \frac{1}{2}\varepsilon \cos(q_1 - q_2)$ Poisson-commute and add up to H . Deduce that F is a first integral, independent of H on a dense open set, for every ε . (iii) Write $F = F_0 + \varepsilon F_1$. Check both conclusions of Lecture 7’s proposition on Poincaré’s mechanism at first order for this F , and show that $dF_0 \wedge dH_0 \neq 0$ off $\mathfrak{P}$. The lecture concludes, for a dense $\mathfrak{P}$, that $dF_0 \wedge dH_0 = 0$ everywhere. Which step of that argument fails here?